

Select Solutions to Introduction to Homotopy Type Theory, Egbert Rijke

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§1.1

1. Give a derivation for the following **element conversion rule**:

$$\frac{\Gamma \vdash A \doteq A' \text{ type} \quad \Gamma \vdash a : A}{\Gamma \vdash a : A'}$$

2. Give a derivation for the following **congruence rule** for element conversion:

$$\frac{\Gamma \vdash A \doteq A' \text{ type} \quad \Gamma \vdash a \doteq b : A}{\Gamma \vdash a \doteq b : A'}$$

Solution #1

$$\text{subst} \frac{\Gamma \vdash a : A \quad \text{var-conv} \frac{\text{sym} \frac{\Gamma \vdash A \doteq A' \text{ type}}{\Gamma \vdash A' \doteq A \text{ type}} \quad \text{var} \frac{}{\Gamma, x : A' \vdash x : A'}}{\Gamma, x : A \vdash x : A'}}{\Gamma \vdash x[a/x] : A'[a/x]}$$

Solution #2